

Team Presentation Script

Meds Consulting LLC

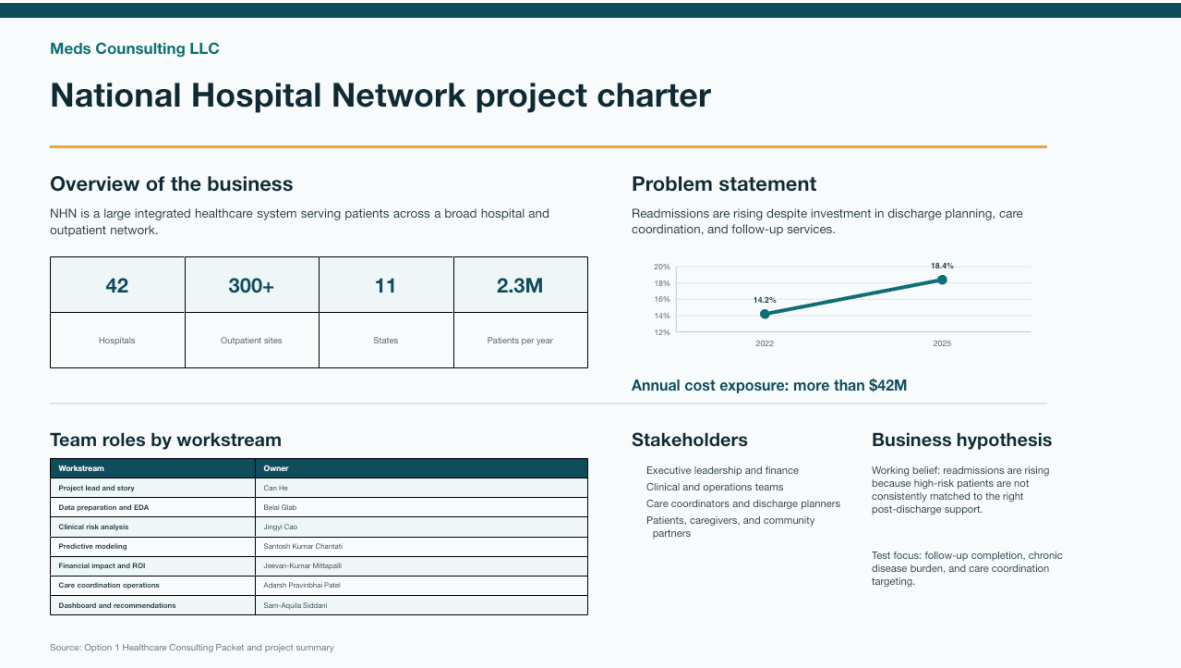
Saturday submission

Total target time: 12-13 minutes, with 2-3 minutes left for transitions or questions.

1. Can He - Project Lead and Story

Slide 1: National Hospital Network project charter

Target time: 1.5 minutes



Slide 1 - National Hospital Network project charter

Good morning everyone. We are Meds Consulting LLC, and today we are presenting our readmission analytics project for National Hospital Network, or NHN.

NHN is a large integrated healthcare system with 42 hospitals, more than 300 outpatient sites, operations across 11 states, and about 2.3 million patients served each year. Because of that scale, even a small increase in readmissions can create a major clinical, operational, and financial challenge.

The business problem is that 30-day readmissions are rising despite investment in discharge planning, care coordination, and follow-up services. The readmission rate increased from 14.2 percent in 2022 to 18.4 percent in 2025, and leadership estimates more than 42 million dollars in annual cost exposure.

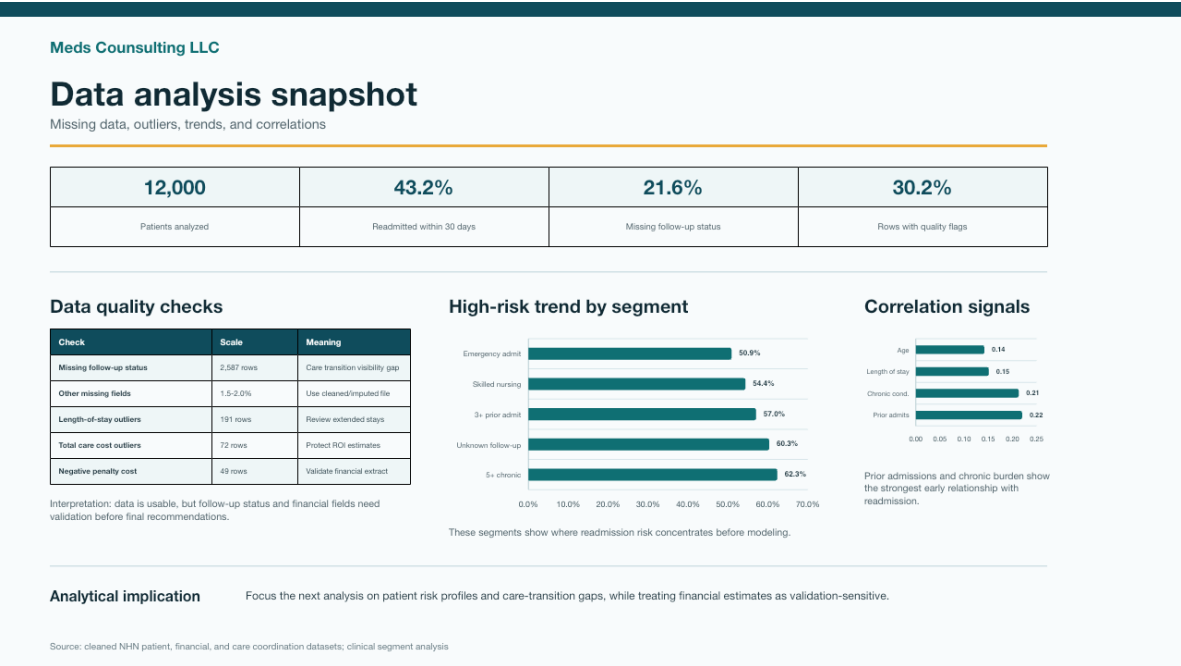
Our working hypothesis is that readmissions are rising because high-risk patients are not consistently matched to the right post-discharge support. We tested this through data analysis, predictive modeling, financial impact analysis, care coordination review, and ethical governance considerations.

I will now hand it over to Belal to walk through the initial data analysis.

2. Belal Glab - Data Preparation and EDA

Slide 2: Data analysis snapshot

Target time: 1.5 minutes



Slide 2 - Data analysis snapshot

Thank you, Can. For the data analysis, we started by checking the quality and structure of the dataset before moving into deeper modeling.

We analyzed 12,000 patients, and the overall 30-day readmission rate in the cleaned data is 43.2 percent. The first important data issue is missing follow-up status. Follow-up status is missing for 2,587 patients, or 21.6 percent of the dataset. This matters because follow-up status is directly connected to the care transition process.

We also checked outliers and quality flags. Length of stay has 191 outlier records, total care cost has 72 outlier records, and there are 49 records with negative penalty cost. These issues do not make the data unusable, but they do tell us that follow-up and financial fields need validation before final decisions.

The key takeaway from this slide is that the data is strong enough to support analysis, but we need to treat follow-up documentation and financial estimates carefully.

Next, Jingyi will explain what the data shows from a clinical risk perspective.

3. Jingyi Cao - Clinical Risk Analysis

Stay on Slide 2: Clinical risk patterns

Target time: 1.5 minutes

Building on the data review, the clinical risk patterns show that readmission risk is not evenly distributed across patients.

The highest-risk groups are clinically intuitive. Patients with five or more chronic conditions have a readmission rate of about 62.3 percent. Patients with unknown follow-up status have a readmission rate of about 60.3 percent. Patients with three or more prior admissions, skilled nursing discharges, and emergency admissions also show elevated readmission rates.

The correlation results support the same direction. Prior admissions and chronic conditions show the strongest positive relationship with readmission, followed by length of stay and age.

From a clinical perspective, this suggests that NHN should not treat all patients as one average group. The opportunity is to identify patients with higher risk before discharge and focus care coordination resources on those groups.

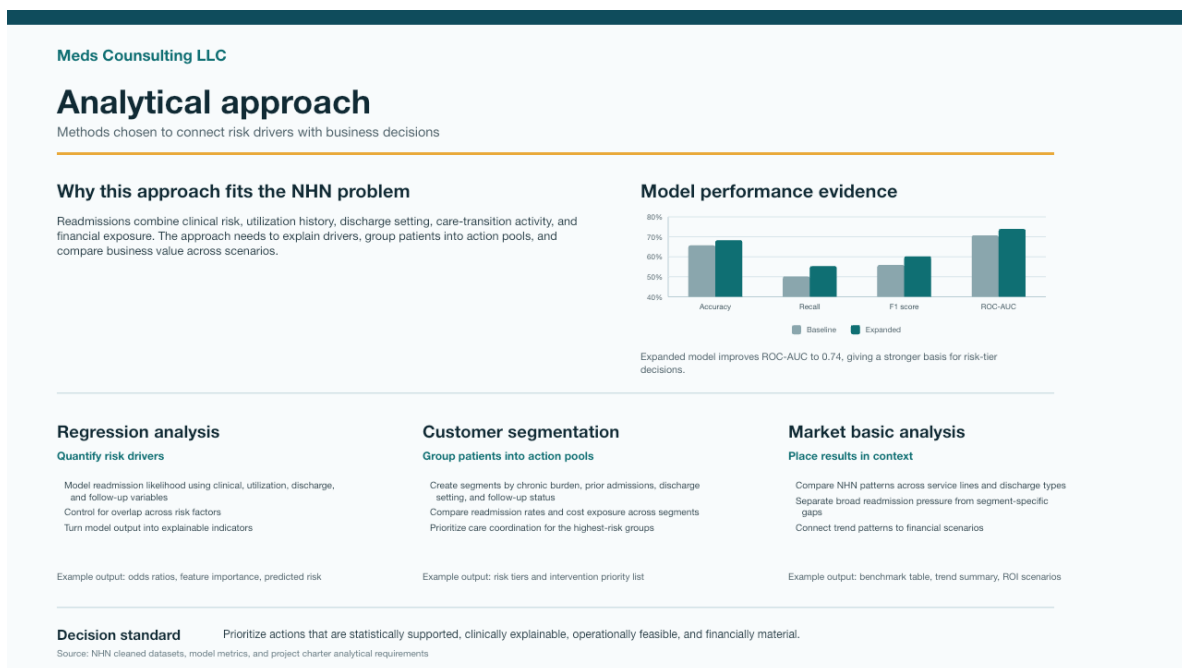
This also creates one unexpected finding: unknown follow-up status is not just a data issue. It may also signal a real care-transition visibility gap.

Now Santosh will explain the analytical approach and modeling logic.

4. Santosh Kumar Chantati - Predictive Modeling

Slide 3: Analytical approach

Target time: 1.5 minutes



Slide 3 - Analytical approach

Thank you, Jingyi. Our analytical approach has three main parts: regression analysis, customer segmentation, and market basic analysis.

First, regression analysis helps us quantify which factors are associated with readmission. This is important because many risk factors overlap. For example, older patients may also have more chronic conditions, more prior admissions, or longer hospital stays. Regression helps us understand which variables still matter after accounting for that overlap.

Second, customer segmentation helps translate the model into action. In this healthcare context, segmentation means grouping patients by chronic burden, prior admissions, discharge setting, and follow-up status. That allows NHN to focus care coordination on the patients most likely to benefit.

Third, market basic analysis gives context. It helps compare patterns across discharge types, care settings, and financial scenarios.

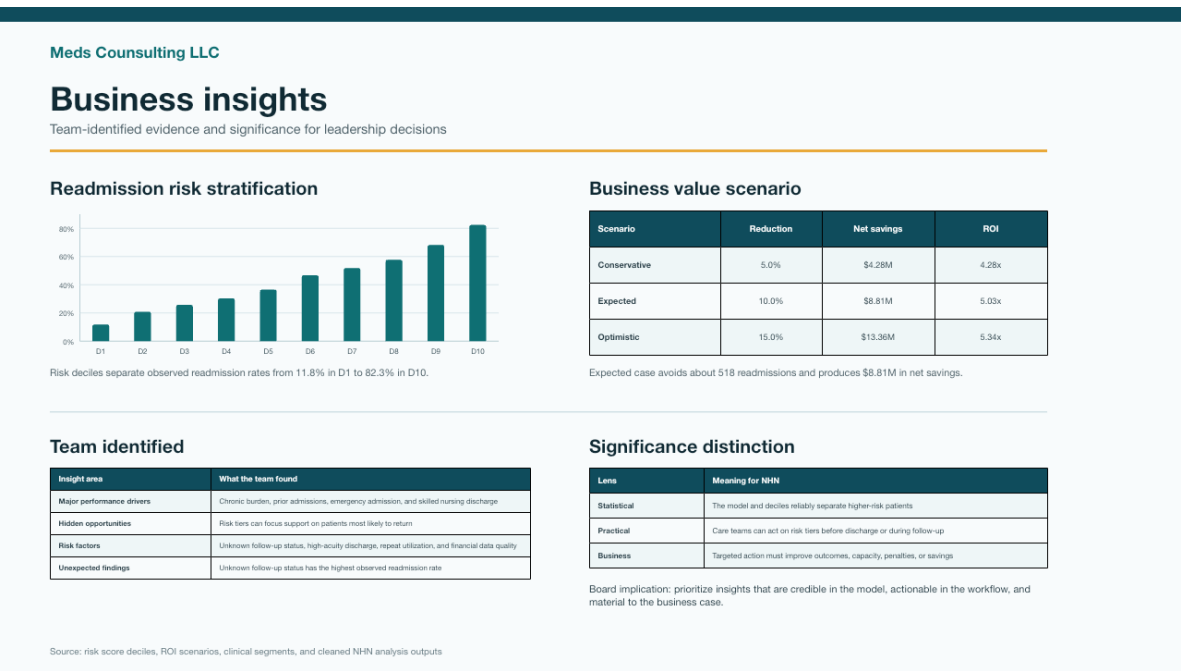
The model performance evidence supports this approach. The expanded clinical model improves ROC-AUC to about 0.74, which gives NHN a stronger basis for risk-tier decisions.

Next, Jeevan will connect these insights to business value and ROI.

5. Jeevan-Kumar Mittapalli - Financial Impact and ROI

Slide 4: Business insights

Target time: 1.5 minutes



Slide 4 - Business insights

Thank you, Santosh. From a business perspective, the main question is whether the analysis can lead to measurable value for NHN.

On Slide 4, the risk decile chart shows that the model meaningfully separates patients by readmission risk. Readmission rates increase from 11.8 percent in the lowest decile to 82.3 percent in the highest decile. This means the model can help NHN identify where intervention is most needed.

The ROI scenario then translates that clinical opportunity into business value. In the expected case, a 10 percent reduction in readmissions avoids about 518 readmissions. The estimated implementation cost is 1.75 million dollars, and the expected net savings are about 8.81 million dollars. That produces an estimated ROI of 5.03 times.

These numbers should be treated as planning estimates, not final budget commitments. Finance should validate the cost, penalty, and intervention assumptions before implementation.

Now Adarsh will explain how these recommendations can be implemented operationally.

6. Adarsh Pravinbhai Patel - Care Coordination Operations

Slide 5: Recommendations and implementation timeline

Target time: 1.5 minutes

Recommendations

Actions, strategy, cost, benefits, timeline, and risks

\$1.75M	518	\$8.81M	5.03x
Estimated implementation cost	Expected avoided readmissions	Expected net savings	Expected ROI

Recommendation roadmap

Action	Recommendation	Timing	Expected benefit
Short term	Pilot pre-discharge risk scoring for emergency and skilled nursing discharges	0-3 months	Earlier targeting of highest-risk patients
Short term	Close follow-up documentation and completion gaps	0-3 months	Lower unknown follow-up status rate
Short term	Approve dashboard KPIs and readmission reduction targets	0-3 months	Consistent executive reporting
Long term	Expand care coordination for chronic disease and repeat-admission groups	3-6 months	Higher intervention completion in high-risk groups
Long term	Track ROI, penalty exposure, and model performance monthly	6-12 months	Sustained savings and model governance

Cost and benefit logic

Expected scenario assumes a 10.0% readmission reduction, \$1.75M implementation cost, and 518 avoided readmissions. Finance should validate assumptions before budget commitment.

Source: recommendation evidence matrix, implementation roadmap, ROI scenarios, and success metrics

Implementation timeline

Phase	Focus
0-3 months	Risk-score pilot, follow-up workflow fix, KPI governance
3-6 months	Targeted care coordination expansion and executive dashboard
6-12 months	Monthly ROI tracking and model refresh

Potential risks

Risk score adoption may stall without clear clinical workflow ownership
Follow-up documentation gaps may hide operational performance issues
ROI estimates depend on validated cost, penalty, and intervention assumptions

Decision ask

Approve a 0-3 month pilot focused on risk scoring, follow-up visibility, and discharge review.

Slide 5 - Recommendations

Thank you, Jeevan. The recommendations are organized into short-term actions and long-term strategy.

In the short term, NHN should pilot pre-discharge readmission risk scoring for emergency admissions and skilled nursing discharges. These groups show elevated risk, so they are a practical starting point. NHN should also close follow-up documentation and completion gaps because unknown follow-up status is both a data quality issue and a potential workflow issue.

The third short-term action is to approve dashboard KPIs and readmission reduction targets. This creates a shared definition of success before the pilot expands.

In the longer term, NHN should expand care coordination for high-risk chronic disease and repeat-admission groups. NHN should also track ROI, penalty exposure, intervention performance, and model performance monthly.

The timeline is staged: 0 to 3 months for pilot and workflow fixes, 3 to 6 months for expansion and dashboard rollout, and 6 to 12 months for ROI tracking and model refresh.

Finally, Sam will cover ethical considerations and responsible use.

7. Sam-Aquila Siddani - Dashboard, Recommendations, and Ethical Considerations

Slide 6: Ethical considerations

Target time: 1.5 minutes

Ethical considerations

Guardrails for fair, private, transparent, and compliant analytics

Fairness	Privacy	Governance	Compliance
Bias checks before use	Minimum necessary data	Clear ownership and audit trail	HIPAA and responsible AI review

Ethical risk areas

Issue	Concern for NHN	Required control
Algorithmic bias	Risk scores may under-serve groups if past care access was unequal	Monitor performance by age, diagnosis, discharge type, and insurance where appropriate
Customer privacy	Patient-level health and financial fields are sensitive	Use minimum necessary data and role-based access
Data governance	Missing follow-up and financial anomalies can distort decisions	Define owners, validation rules, and change logs
Transparency	Clinicians need to understand why a patient is high risk	Use explainable model outputs and plain-language score guidance
Regulatory compliance	Analytics must align with HIPAA and internal policy	Review data use, access, retention, and reporting with compliance leaders
Responsible AI	Predictions should support judgment, not replace it	Keep human review before care decisions and budget commitments

Decision standard

Use the model as a decision-support tool only when fairness checks, privacy controls, data governance, transparency, and compliance review are in place.

Source: project requirements, NHN data quality findings, risk model workflow, and healthcare compliance context

Operating principles

Principle	How to apply it
Fair	Test whether model errors differ across patient groups
Private	Limit data access to teams with a clear care or reporting need
Explainable	Show the key factors behind each high-risk flag
Accountable	Assign owners for model refresh, dashboard definitions, and issue escalation

Governance workflow

Validate data quality before model refresh or ROI reporting
Document model purpose, limitations, and approved user groups
Escalate fairness, privacy, or compliance concerns before scaling

Practical implication Care teams keep final judgment over patient outreach and discharge decisions.

Slide 6 - Ethical considerations

Thank you, Adarsh. The final slide focuses on ethical considerations because a readmission model affects patient care decisions.

The first issue is algorithmic bias. If historical data reflects unequal access to care or documentation gaps, the model may perform differently across patient groups. NHN should monitor performance by relevant groups such as age, diagnosis, discharge type, and insurance where appropriate.

The second issue is patient privacy. The analysis uses sensitive patient-level health, utilization, care coordination, and financial data. Access should follow the minimum necessary principle, with role-based controls.

The third issue is data governance. Missing follow-up values and financial anomalies need clear ownership, validation rules, refresh timing, and change logs.

The fourth issue is transparency. Clinicians need to understand why a patient is flagged as high risk. The model should show key drivers in plain language.

Finally, responsible AI means the model should support human judgment, not replace it. Care teams should keep final decision-making authority over outreach and discharge planning.

That concludes our presentation. Our recommendation is to start with a controlled pilot, validate the results, and scale only with strong governance in place.